Interim Analysis

Ciaran Evans

Motivation: Stopping early?

Observe data $X_1, ..., X_n$ over n months, with single X_i collected each month. Assume that $X_1, ..., X_n \overset{iid}{\sim} N(\mu, 1)$

Want to test $H_0 : \mu = 0$ vs. $H_A : \mu > 0$. Let $Z_k = \frac{1}{\sqrt{k}} \sum_{i=1}^{k} X_i$.

Decision rule: Reject H_0 if $Z_k > c_1$ or $Z_n > c_2$.

Bonferroni approach:

$P(Z_k > c_1 \text{ or } Z_n > c_2) \leq P(Z_k > c_1) + P(Z_n > c_2)$

choose c_1, c_2 st $P(Z_k > c_1) = P(Z_n > c_2) = \frac{\alpha}{2}$

e.g. $\alpha = 0.05 \Rightarrow c_1, c_2 = 1.96$

General union bound:

c_1, c_2 st $P(Z_k > c_1) = \alpha_1$ $\quad P(Z_n > c_2) = \alpha_2$

$\alpha_1 + \alpha_2 = \alpha$

Exact probability of a type I error

$$P(Z_k > c_1 \cup Z_n > c_2) = P(Z_k > c_1) + P(Z_k < c_1, Z_n > c_2)$$

$Z_k \sim N(0,1)$ under H_0 $\Rightarrow$ $P(Z_k > c_1) = 1 - \Phi(c_1)$

$$P(Z_k < c_1, Z_n > c_2) = \int_{-\infty}^{c_1} P(Z_n > c_2 \mid Z_k = t)\, f_{Z_k}(t)\, dt$$

$$P(Z_n > c_2 \mid Z_k = t) = P\left(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i > c_2 \,\middle|\, \frac{1}{\sqrt{k}} \sum_{i=1}^{k} X_i = t\right)$$

$$= P\left(\frac{1}{\sqrt{n}} \left(\sum_{i=k+1}^{n} X_i + \sqrt{k}\, t\right) > c_2\right)$$

$$= P\left(\sum_{i=k+1}^{n} X_i > \sqrt{n}\, c_2 - \sqrt{k}\, t\right)$$

$$= P\left(\frac{1}{\sqrt{n-k}} \sum_{i=k+1}^{n} X_i > \frac{1}{\sqrt{n-k}} \left(\sqrt{n}\, c_2 - \sqrt{k}\, t\right)\right)$$

$$\overset{(H_0)}{=} 1 - \Phi\left(\frac{1}{\sqrt{n-k}} \left(\sqrt{n}\, c_2 - \sqrt{k}\, t\right)\right)$$

$$\Rightarrow P(Z_k > c_1, Z_n > c_2) \overset{(H_0)}{=} \int_{-\infty}^{c_1} \left(1 - \Phi\left(\frac{1}{\sqrt{n-k}} \left(\sqrt{n}\, c_2 - \sqrt{k}\, t\right)\right)\right) \phi(t)\, dt$$

Pocock approach

$$(1-\Phi(c_1)) + \int_{-\infty}^{c_1} \left(1-\Phi\left(\frac{1}{\sqrt{n-k}}(\sqrt{n}\,c_2-\sqrt{k}\,t)\right)\right)\phi(t)\,dt$$

$$P(Z_k > c_1 \cup Z_n > c_2) = P(Z_k > c_1) + P(Z_k < c_1, Z_n > c_2)$$

Pocock approach:

Choose $c_1 = c_2 = c$ (same threshold at each stage of the analysis)

<u>Pocock in general:</u>

1. Specify total desired # of analyses, R
2. Do the analyses at regularly spaced intervals. i.e. collect $m = \frac{n}{R}$ observations in each interval, and test hypotheses after $m, 2m, 3m, \ldots, n$ observations are collected

Disadvantage: makes it harder to reject at final stage when we have most data!

O'Brien-Fleming (OBF) approach

Motivation: Put most of α into the final analysis, so significance level of final analysis is close to that of a single test procedure

Setup: Observe $X_1, ..., X_n$. Perform R total analyses: interim analyses at times $k_1, k_2, ..., k_{R-1} < n$, and final analysis at time n.

<u>OBF method</u>: Let c_R be threshold for final analysis

$$c_1 = \sqrt{R} \cdot c_R$$

$$c_2 = \sqrt{\frac{R}{2}} \cdot c_R$$

$$\vdots$$

$$c_{R-1} = \sqrt{\frac{R}{R-1}} \cdot c_R$$

makes interim thresholds very stringent, whereas final threshold is close to threshold for a single test

Example

$$P(Z_k > c_1 \cup Z_n > c_2 | \mu = 0) =$$

$$(1 - \Phi(c_1)) + \int_{-\infty}^{c_1} \left(1 - \Phi\left(\frac{1}{\sqrt{n-k}}(\sqrt{n}c_2 - \sqrt{k}t)\right)\right) \phi(t)dt$$

Example: $R = 2$ (so $k = n/2$), and $\alpha = 0.05$

<u>Pocock</u>: $c_1 = c_2 = c = 1.875$

$P(Z_k > c_1) \approx 0.03$ $\quad$ $P(Z_k < c_1, Z_n > c_2) \approx 0.02$

<u>OBF</u>: $c_1 = \sqrt{2}\, c_2 \Rightarrow c_1 = 2.373$ (very stringent)

$c_2 = 1.678$ (close to single-test cutoff 1.645)

$P(Z_k > 2.373) \approx 0.009$

$P(Z_k < 2.373, Z_n > 1.678) \approx 0.041$

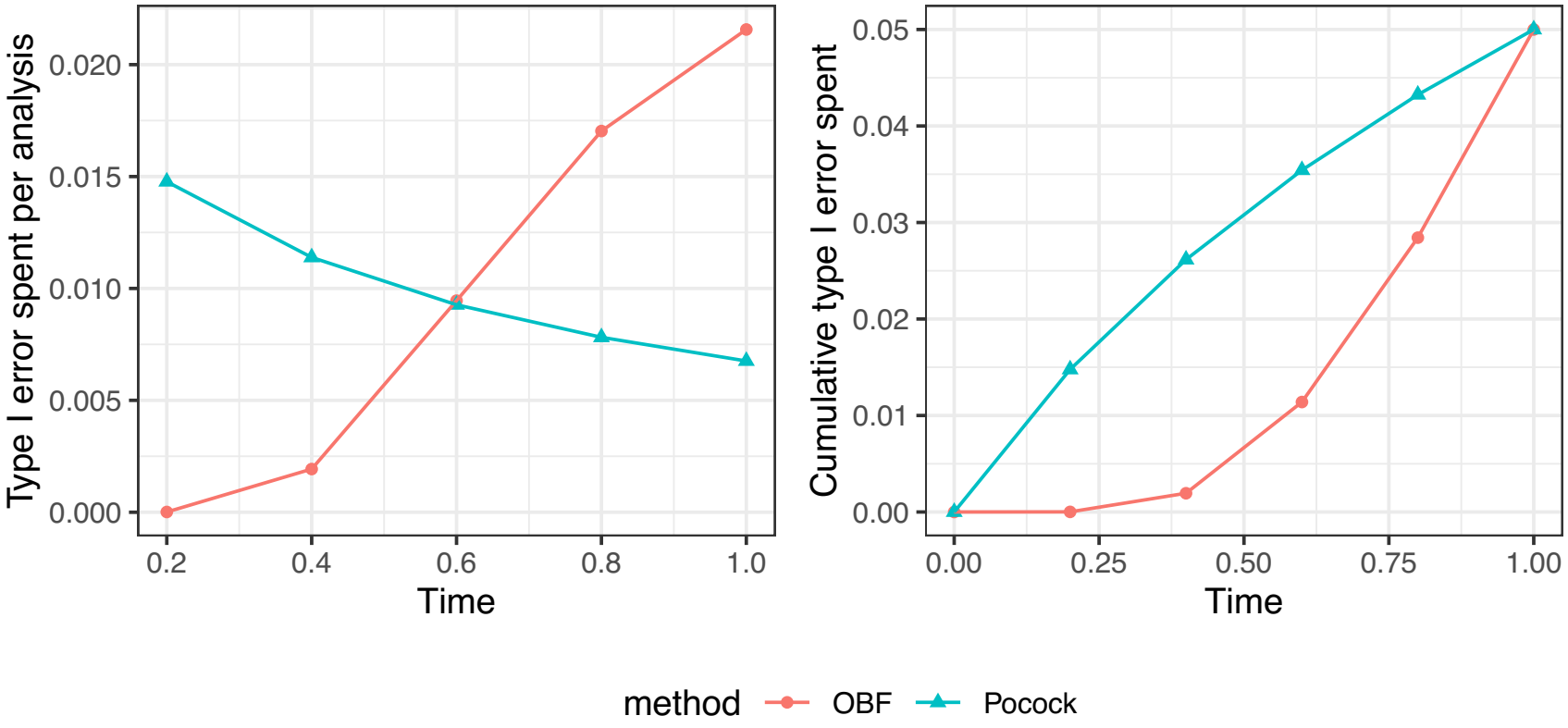
Drawbacks

There are several drawbacks to the Pocock and OBF approaches:

1. Have to prespecify number of analyses, and when you will conduct them.
2. Published Pocock/OBF bounds assume evenly-spaced tests. (Can calculate your own boundaries for non-evenly spaced tests, but that isn't as easy as looking up existing boundaries)

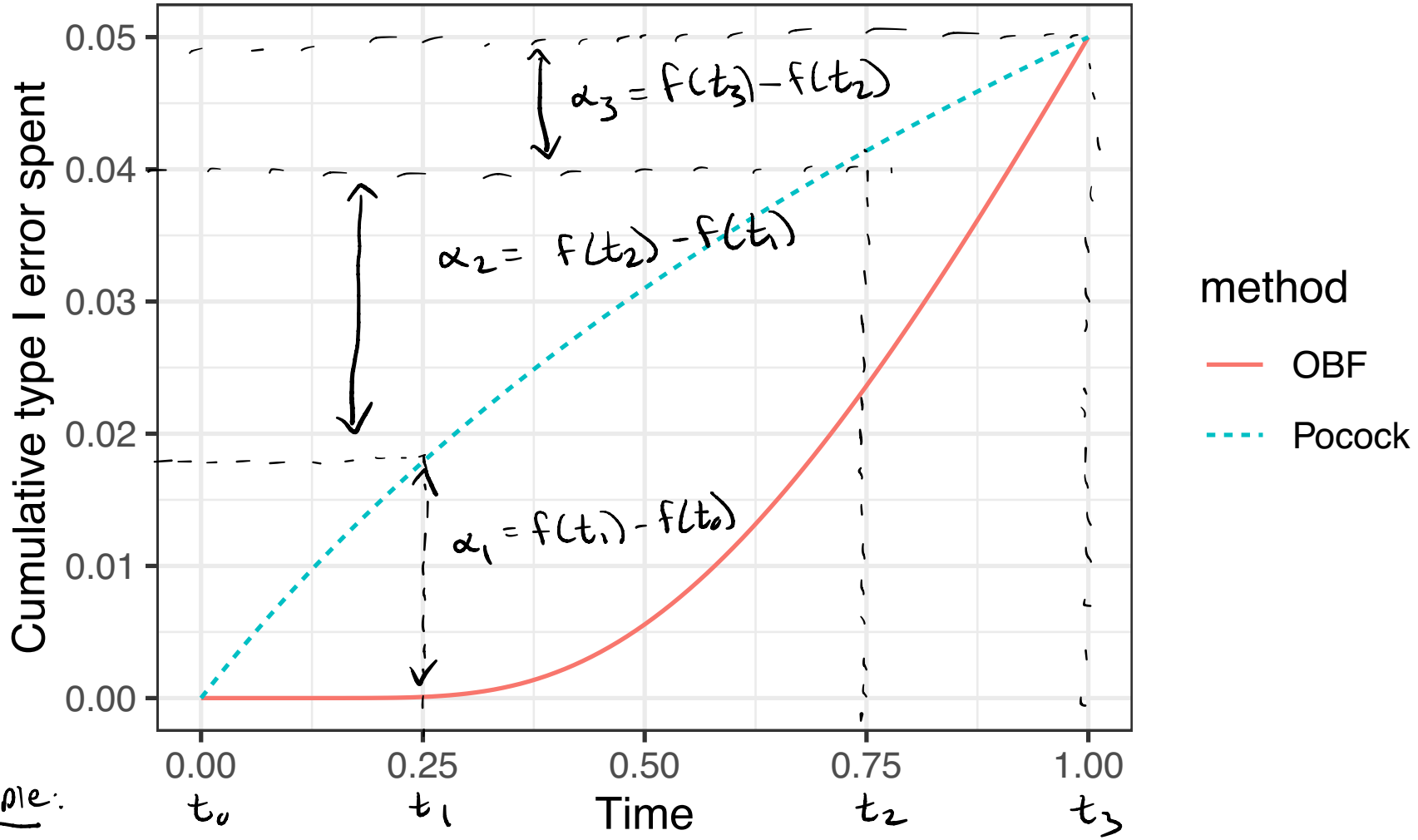
α spending

Suppose we wish to do $R = 5$ total analyses, and we want an overall type I error rate of $\alpha = 0.05$. OBF and Pocock bounds specify the amount of type I error "budget" that we use for each analysis:

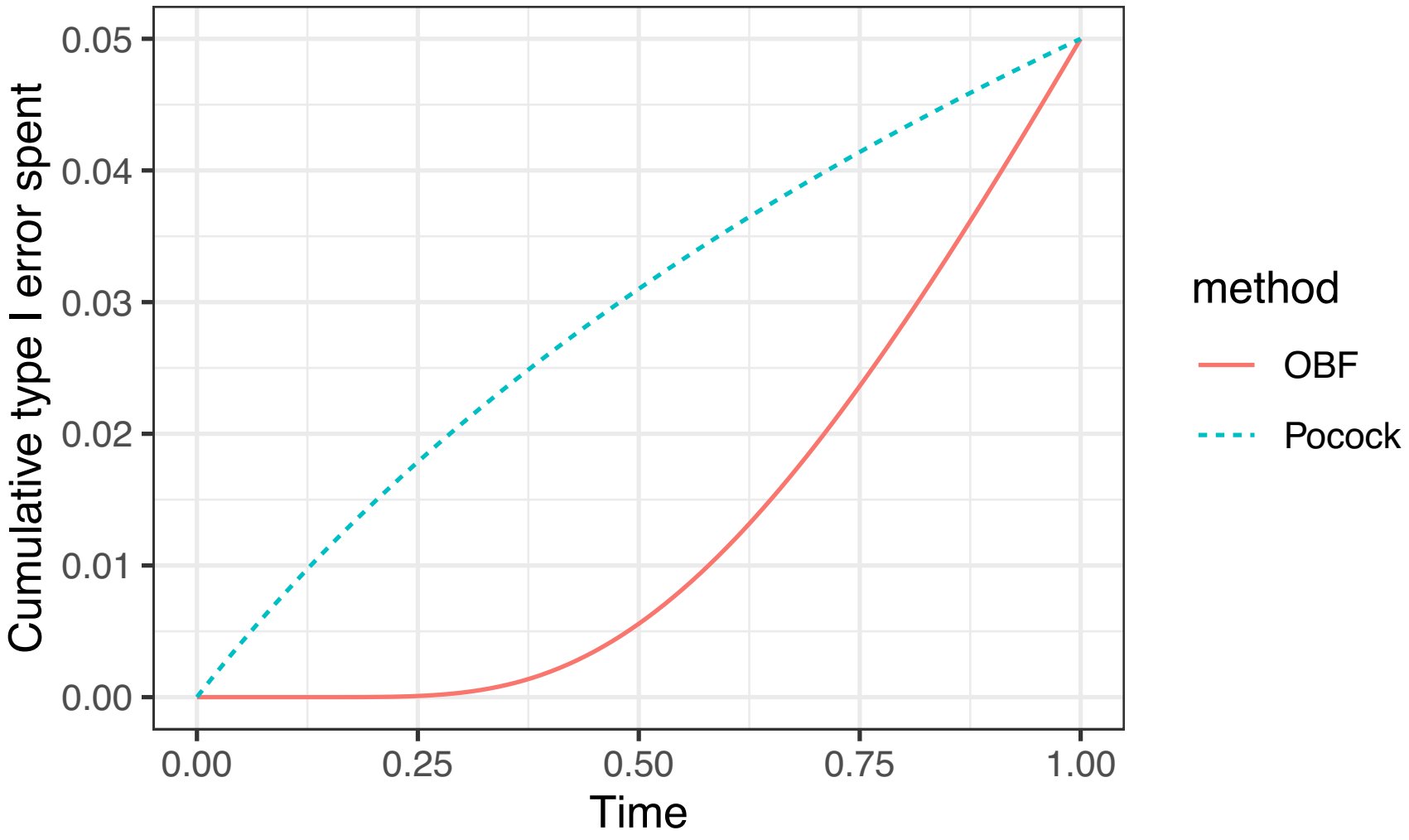


α spending functions

- Non-decreasing function $f(t) : [0, 1] \to [0, \alpha]$ with $f(0) = 0$ and $f(1) = \alpha$, which gives **total** type I error spent by time t
- Conduct analyses at times $0 < t_1 < t_2 < \cdots t_R = 1$
- Portion of α spent on analysis t_i is $f(t_i) - f(t_{i-1})$



Example:

α spending functions

Advantage: Don't have to specify number of interim analyses, or when the analyses will be performed, ahead of time